AJAY KUMAR GARG ENGINEERING COLLEGE, GHAZIABAD

Department of Computer Science and Engineering

VII Semester Project: SECOND PROJECT VIVA (12–13 October 2026)

VIVA-II SELF-EVALUATION FORM

Field	Details
Group No. / Code	Group 15 (2023-2027/CSE-DS/1/G15)
Project Title	AI-Driven Multimodal Sentiment Analysis for Mental Health Monitoring on Social Media
Guide	Dr. Avdhesh Gupta
Members (Roll No.)	Kartikey Tiwari (2300271540063) Akshat Sharma (2300271540018) Karan Sharma (2300271540062) Lalit Sharma (2300271540066)
Domain	AI/ML: NLP, Computer Vision, Multimodal Learning

A. Self-Assessment Against the Viva-II Criteria

#	Criterion	What the group has completed (evidence)	Status	Self-score
1	Research Progress	Problem statement, primary RQ + 4 secondary RQs, 6 objectives defined. 5 of 6 objectives completed; dissemination (O6) in progress. Evidence: Extended Synopsis Ch. 1, Table 1.1.	Completed	
2	Research Design	Experimental comparative design: same split, seed and metrics for every model. Architecture: text branch (TF-IDF / DistilBERT), image branch (CNN / ResNet18), concat-MLP and late fusion. Each choice justified against CPU-only constraints and the literature (Ch. 3, Tables 3.2–3.3).	Completed	
3	Implementation	Modular Python codebase: data, preprocessing, text, image, fusion, evaluation, explainability. Streamlit demo + research dashboard (screenshots, live demo). Datasets: MVSA-Single (4,511), Memotion (3,500). Tools: PyTorch, Transformers, scikit-learn. 32/32 automated tests pass. 7 technical challenges documented with resolutions (Ch. 4).	Completed	
4	Results & Validation	Study 1 (MVSA): fusion 64.4% acc / 56.3% macro-F1 vs. best text 61.6% / 54.6% and best image 52.7% / 42.8%. Modality-conflict analysis: the fusion gain is concentrated where text and image disagree (57.1% vs. 50.0%). Study 2 (memes): blur ablation removes 10.9 pp of fusion accuracy, so the gain comes mainly from embedded-text leakage. Leakage audit, confusion matrices, explainability (Ch. 5).	Completed (multi-seed pending)	
5	Next Plan	Multi-seed + significance tests (Oct); OCR-masking ablation and full Memotion run (Nov); attention fusion as a stretch goal (Nov–Dec); manuscript (Dec); submission (Jan 2027); final report (Feb–Mar 2027). Gantt chart: Ch. 6.	Planned with timeline	
6	Presentation & Q&A	Slide deck (PRESENTATION.pptx), IEEE-format paper drafts (2 studies), viva Q&A preparation (VIVA_PREPARATION.md). Every member can explain their module and the modality-conflict and blur findings.	Ready	

B. Deliverables Brought to the Viva

Deliverable	Status
Extended Synopsis with Research Design and Implementation sections (signed by Guide)	Prepared: Extended_Synopsis_Viva2.docx / .pdf
Viva-II Self-Evaluation Form (signed by Guide)	This form
Presentation slides	PRESENTATION.pptx

Working demonstration	streamlit run app/streamlit_app.py
Research paper drafts (IEEE format)	research_paper/paper_IEEE.pdf, research_paper_v2/paper_IEEE.pdf

C. Individual Contribution

Member	Contribution so far
Kartikey Tiwari	Literature review, dataset acquisition (MVSA mirror verification, Memotion), integration
Akshat Sharma	Text preprocessing, TF-IDF and DistilBERT models
Karan Sharma	CNN / ResNet18 image models, fusion models, evaluation and blur ablation
Lalit Sharma	Documentation, research paper, presentation, testing

Challenges faced: dead official dataset link (resolved via a verified mirror); CPU-only, 8 GB RAM (resolved with frozen encoders and cached embeddings); class imbalance (class weighting, macro-F1); no public multimodal mental-health-labelled dataset (sentiment used as a non-clinical proxy).

Signatures of Group Members:

Kartikey Tiwari	Akshat Sharma	Karan Sharma	Lalit Sharma

Remarks of Guide:

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Signature of Guide (Dr. Avdhesh Gupta): _____ Date: _____